Requirements:

1. Create a complete backtesting system that implements:
    - Proper data preparation with handling of missing values and adjustments
    - Realistic simulation of trading costs (commissions, slippage)
    - Accurate position sizing and trade execution
    - Comprehensive performance metrics (Sharpe, Calmar, drawdown, win rate)
2. Implement a moving average crossover strategy with the following parameters to optimize:
    - Short-term moving average period (test range: 5-50 days)
    - Long-term moving average period (test range: 50-200 days)
    - Optional position sizing parameter (fixed percentage vs. volatility-adjusted)
3. Develop a Walk-Forward Analysis framework that:
    - Divides your data into multiple training/testing segments
    - Optimizes parameters on each training segment
    - Tests the optimized parameters on the subsequent testing segment
    - Analyzes parameter stability across different time periods
    - Calculates Sharpe Ratio retention between training and testing periods
4. Apply your system to at least 3 different stocks or ETFs with diverse characteristics:
    - A technology stock (e.g., AAPL, MSFT)
    - A broad market index ETF (e.g., SPY, QQQ)